

The Ultimate Web Development Handbook



Introduction to HTML

What is HTML?

HTML stands for **HyperText Markup Language**.

It is the standard language used to create and structure webpages.

Web browsers (like Chrome, Firefox, Safari) read HTML code and display it as websites.

Why Learn HTML?

- It's the **foundation** of all websites.
 - You need HTML to build pages before adding CSS (styling) or JavaScript (functionality).
 - Knowing HTML helps you understand how websites work behind the scenes.
-

Basic Terminology

- **Element**: A piece of content in a webpage (like a paragraph, heading, or image).
 - **Tag**: Special keywords inside angle brackets like `<p>` or `<h1>` that define elements.
 - **Attribute**: Extra information added to tags, like `href` in links or `src` in images.
-

Basic Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Page</title>
  </head>
  <body>
    <h1>Hello, world!</h1>
    <p>This is my first HTML page.</p>
  </body>
</html>
```

Explanation:

- `<!DOCTYPE html>` tells the browser this is an HTML5 document.
- `<html>` is the root of the HTML page.
- `<head>` contains information about the page (not shown on screen).
- `<title>` sets the title seen on the browser tab.
- `<body>` contains everything visible on the webpage.

Key Points

- HTML is made up of **tags**.
- Tags usually come in **pairs**: an opening tag `<p>` and a closing tag `</p>` .
- The content goes **between** the tags.
- Indentation helps make code easier to read, but it's not required.

Basic HTML Structure

What is the Basic Structure of an HTML Page?

Every HTML document follows a basic structure. This structure tells the browser how to read and display the content.

Template of a Basic HTML Page

```
<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head>
  <body>
    <!-- Your content goes here -->
  </body>
</html>
```

Explanation of Each Part

1. `<!DOCTYPE html>`

- Declares that this is an HTML5 document.
- Must be the first line in the file.

2. `<html>...</html>`

- The root element of the page.
- Wraps all the content of your HTML document.

3. `<head>...</head>`

- Contains meta-information **about** the page.
- This can include:
 - The page `<title>`
 - Links to CSS files
 - Meta tags (like keywords, description, etc.)

4. `<title>...</title>`

- Sets the name shown on the browser **tab**.

5. `<body>...</body>`

- Contains everything visible on the page.
- You'll place text, images, links, forms, etc. here.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Web Page</title>
  </head>
  <body>
    <h1>Welcome!</h1>
    <p>This is a simple HTML page with basic structure.</p>
  </body>
</html>
```

Tips

- Always start with `<!DOCTYPE html>` .
- Make sure `<html>` , `<head>` , and `<body>` are properly opened and closed.
- Use indentation to keep your code clean and readable.

Headings and Paragraphs

Headings in HTML

Headings help you organize content into sections.

HTML provides **6 levels** of headings:

- `<h1>` – Main heading (biggest)
- `<h2>` – Subheading
- `<h3>` – Smaller subheading
- `<h4>` , `<h5>` , `<h6>` – Even smaller headings

Example of Headings

```
<h1>This is a Heading 1</h1>
<h2>This is a Heading 2</h2>
<h3>This is a Heading 3</h3>
<h4>This is a Heading 4</h4>
<h5>This is a Heading 5</h5>
<h6>This is a Heading 6</h6>
```

Tip:

- Use **only one** `<h1>` per page (usually for the page title).
 - Use headings to **structure your content**, not to make text look big (that's CSS's job).
-

Paragraphs in HTML

Paragraphs are written using the `<p>` tag.

Example:

```
<p>This is a paragraph. It can contain multiple sentences of text.</p>
```

Notes:

- Browsers automatically add **space** before and after each paragraph.
- You don't need to press Enter manually for new lines. Use a new `<p>` tag instead.

Line Breaks

If you want to break a line **without starting a new paragraph**, use the `<br>` tag.

Example:

```
<p>This is line one.<br>This is line two.</p>
```

`<br>` is a **self-closing tag**, which means it doesn't need a closing `</br>`.

Complete Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>Headings and Paragraphs</title>
  </head>
  <body>
```

```
<h1>My Blog</h1>
<h2>Introduction</h2>
<p>Welcome to my first HTML blog post!</p>

<h2>Why I Love Coding</h2>
<p>Coding lets you build websites, apps, and games.<br>It's fun and creative!</p>
</body>
</html>
```